

Enrolment No:.....

Name of Student.....



BENNETT
UNIVERSITY
THE TIMES GROUP

NAAC
GRADE A+
ACCREDITED UNIVERSITY

End Semester Examinations, May 2026

Even Semester 2025-26

B.Tech./BCA/BBA/B.Tech.- M.Tech./BBA-MBA/B.A.– 2nd Semester

UVAC1003 - ENVIRONMENT AND SUSTAINABILITY

Time: 2hrs.

Max Marks: 40

Note: *Student must attempt 5 Questions in total. First question is compulsory, and student must attempt question(s) from each section. Marks are indicated against each question.*

PART-A

Q.1 This part has 8 questions out of which student must attempt any 6 questions.

- ✓ (a) Explain the concepts of in-situ and ex-situ conservation. [CO-1] [L-2] [1+1=2]
- ✓ (b) Define sustainability and list its pillars. [CO-1] [L-2] [1+1=2]
- (c) Define PM₁₀ and PM_{2.5} and state which particulate matter is most harmful to human health. [CO-3] [L-2] [1+1=2]
- (d) Name the radioactive elements responsible for thyroid cancer and bone marrow cancer. [CO-3] [L-2] [1+1=2]
- (e) Describe the main characteristics of the Tundra biome in terms of climate and vegetation. [CO-1] [L-2] [1+1=2]
- (f) Explain the role of CPCB and SPCB in pollution control. [CO-4] [L-2] [1+1=2]
- ✓ (g) Define the objectives of Kyoto and Montreal protocol. [CO-4] [L-2] [1+1=2]
- (h) What is meant by dB(A) Leq? State the permissible noise levels in residential areas during daytime. [CO-3] [L-2] [1+1=2]

2*6=12 Marks

PART-B

This part has three questions out of which student must attempt any two questions.

- ✓ Q.2 Explain the causes and consequences of water pollution with reference to BOD, pathogenic organisms, and nutrient-induced eutrophication. [CO-3] [L-2] [2+2+2=6]

Q.3 Define wetlands and describe their classification. Explain any two significances of wetland restoration. [CO-1] [L-2] [2+2+2=6]

Q.4 Briefly explain three different scopes of GHG emissions. A university campus generates 120 kg of food waste per day, which is sent to a landfill. The emission factor for food waste disposed in landfills is 2.5 kg CO₂-equivalent per kg of waste. Calculate the daily carbon footprint due to food waste disposal. [CO-2] [L-4] [3+3=6]

6*2=12 Marks

PART-C

Q.5 Differentiate between the greenhouse effect and ozone layer depletion, and analyse their causes, mechanism, and environmental impacts. [CO-2] [L-5] [2+2+2+2=8]

OR

A metropolitan city is experiencing rising temperatures, irregular rainfall patterns, and increased frequency of extreme weather events. Analyse how these changes are linked to climate change and evaluate their impacts on urban infrastructure, water resources, and public health. Suggest appropriate mitigation and adaptation measures.

[CO-2] [L-5] [2+2+2+2=8]

Q.6 Differentiate between composting and anaerobic digestion (biomethanation) based on their processes, end products, advantages, and limitations.

[CO-3] [L-6] [2+2+2+2=8]

OR

A packaging company is evaluating the environmental impact of replacing conventional plastic bottles with biodegradable alternatives. Using this scenario, analyse the four phases of Life Cycle Assessment (LCA). Explain how it helps in making a sustainable decision.

[CO-3] [L-6] [2+2+2+2=8]

8*2=16 Marks
